

4 Calcium

Calcium supports cell walls, membranes, root tips, and fruit tissues. Uptake depends on active roots and water flow, so soil abundance alone may not assure delivery.

Why it matters

- **Structure in the plant.** Calcium helps stabilize cell walls and membranes and is important in growing root and shoot tissue.
- **Limited movement.** Calcium does not readily move from older tissue to new tissue or fruit; continued root uptake and water movement matter.

Tentative leaf guide

Measure	Interpretation note
Tentative avocado range	UC guidance lists less than 0.5 percent as deficient, 1.0 to 3.0 percent as adequate, and more than 4.0 percent as excess.
Tissue caution	A leaf value does not directly measure calcium delivery to fruit or identify a cause of fruit-quality problems.

Main soil and environmental controls

Factor	What it can mean in the root zone
pH and parent material	Acid sandy soils may lose Ca by leaching. Calcareous high-pH soils may contain abundant Ca while iron, zinc, manganese, or P remain poorly available.
Organic matter and CEC	Clay and organic matter hold Ca on exchange sites and help create stable aggregates.
Sodium and structure	Where sodium disperses clay, a soluble Ca source such as gypsum may help only when the diagnosis, rate, water, and drainage are appropriate.
Water and roots	Drought, waterlogging, salinity, poor transpiration, and damaged feeder roots can limit Ca delivery even when soil tests are high.
Cation competition	High K, Mg, ammonium, or sodium can change uptake balance. A single universal Ca to Mg to K ratio is not a reliable target.

Field clues and checks

- **Separate nutrient need from soil repair.** Lime changes acidity; gypsum supplies soluble Ca and sulfate without the same liming effect. Each requires a defined purpose.
- **Inspect irrigation and roots.** Check wetting depth, ponding, salinity, sodium hazard, feeder-root condition, and crop water stress.
- **Track the cation set.** Interpret Ca with K, Mg, Na, CEC, pH, and the actual irrigation-water composition.

DST interpretation Phase 1 highlighted K by Ca and Ca by Mg as context variables. These signals support joint diagnosis of cations and roots, not a fixed ratio or automatic amendment.